[Laurent series of rational polynomial.

Let $f(z)$ and $g(z)$ be polynomials over $\mathbb{C}$, relatively prime

Consider the fraction $\dfrac{g(z)}{f(z)}$.

[Silverman, Exercise in sec 9.1.

For $0<|a|<|b|$, expand $f(z)=\dfrac{1}{[z-a]\cdot[z-b]}$

Then there are five valid Laurent expansions:

$$\frac{1}{(z-a)(z-b)}=\frac{1}{b-a}\cdot\left(\frac{1}{z-b}-\frac{1}{z-a}\right)=\frac{1}{b-a}\left(\frac{1}{a-z}-\frac{1}{b-z}\right)$$

$$\overset{(1)}{=}\frac{1}{b-a}\cdot\left(\frac{1}{a}\cdot\frac{1}{1-(z/a)}-\frac{1}{b}\cdot\frac{1}{1-(z/b)}\right)$$

$$=\frac{1}{b-a}\cdot\left(\frac{1}{a}\cdot\sum_{n=0}^{\infty}\left(\frac{z}{a}\right)^n-\frac{1}{b}\sum_{n=0}^{\infty}\left(\frac{z}{b}\right)^n\right)$$

$$=\frac{1}{b-a}\cdot\left(\sum_{n=0}^{\infty}\left(\frac{1}{a^{n+1}}-\frac{1}{b^{n+1}}\right)z^n\right)\qquad |z|<|a|$$

$$\overset{(2)}{=}\frac{1}{b-a}\cdot\left(\frac{1}{z}\cdot\frac{1}{\left(\frac{a}{z}-1\right)}-\frac{1}{b}\cdot\frac{1}{1-(z/b)}\right)$$

$$=\frac{1}{b-a}\left(-\frac{1}{z}\cdot\sum_{n=0}^{\infty}a^n\cdot z^{-n}-\frac{1}{b}\cdot\sum_{n=0}^{\infty}\left(\frac{z}{b}\right)^n\right)$$

$$=\frac{1}{b-a}\left(-\sum_{n=1}^{\infty}a^{n-1}\cdot z^{-n}-\sum_{n=0}^{\infty}\frac{1}{b^{n+1}}z^n\right)\qquad |a|<|z|<|b|.$$

$$\overset{(3)}{=}\frac{1}{b-a}\cdot\left(\frac{1}{z}\cdot\left(\frac{1}{1-(b/z)}-\frac{1}{1-(a/z)}\right)\right)$$

$$=\frac{1}{b-a}\cdot\frac{1}{z}\cdot\left[\sum_{n=0}^{\infty}\left(\frac{b}{z}\right)^n-\left(\frac{a}{z}\right)^n\right]$$

$$=\frac{1}{b-a}\cdot\sum_{n=1}^{\infty}\left(b^{n-1}-a^{n-1}\right)\cdot z^{-n}\qquad |z|>|b|.$$

$$\overset{(4)}{=}\frac{1}{b-a}\left(\frac{1}{a-b+(z-a)}-\frac{1}{z-a}\right)$$

$$=\frac{1}{b-a}\left(\frac{1}{a-b}\cdot\frac{1}{1-(z-a)/(b-a)}-\frac{1}{z-a}\right)$$

$$=\frac{1}{b-a}\cdot\left(\frac{1}{a-b}\cdot\sum_{n=0}^{\infty}\left(\frac{z-a}{b-a}\right)^n-\frac{1}{z-a}\right)\qquad 0<|z-a|<|a-b|.$$